

SPECTRAL SYNTHESIS OF HYPERSURFACES IN $\mathbb{R}^n$

SHANTANU DEODHAR

Abstract: Agranovsky and Narayanan proved that if a function $f \in L^p(\mathbb{R}^n)$ for $p \leq \frac{2n}{d}$ and its Fourier transform is supported on a d -dimensional sub-manifold, then $f \equiv 0$. We address the natural refinement of this problem for a smooth, compact hypersurface $S \subset \mathbb{R}^n$: What is the largest exponent p such that any $f \in L^p(\mathbb{R}^n)$, with $\text{supp } \hat{f} \subset S$, must be identically zero?

We improve upon the Agranovsky-Narayanan bound by a covering argument that tiles the hypersurface with anisotropic rectangles adapted to its local curvature profile. I will also outline how this approach may extend to higher-codimensional manifolds and highlight the geometric and analytic challenges that arise in attempting such a generalization.